Corn Cultivation and Climatic Conditions

ITM University Gwalior · School of Computing and Intelligent Systems

· Project-Based Learning (PBL) Mini Project Report

Course	Data Analytics using Python (CSL-0312)
Program	B.Tech. CSE (AIML), Semester III, Section E
Session	July–Dec 2026
Domain	Corn Suitability Modeling
Team Leader	Ch. Mohith (BETN1AI25149)
Team Members	Ayan Khan (BETN1AI25138), Mohammad Jubaid (BETN1AI25080)
Faculty Mentor	Dr. Sanjay Jain, Professor, Dept. of CSE
Date of Submission	21/10/2026

1. Introduction & Problem Statement

Corn is one of the world's most important crops, used for food, animal feed and fuel (ethanol). Illinois is one of the biggest corn-producing states in the USA. Corn depends strongly on weather: heat and drought in July, when the plant is pollinating, can cut yield sharply, as happened across the US Corn Belt in 2012.

Farmers, insurers and planners need to know which conditions suit corn and how much weather variation puts production at risk.

Scope (as advised by our faculty mentor): one state (Illinois), ready-made CSV files instead of raw maps, and simple analysis in Python (pandas, Matplotlib, Seaborn, SciPy): cleaning, exploration, charts and basic statistics.

2. Dataset Description & Source

Four public files were combined into one table.

File	Source	Rows	What it contains
01_raw_usda_nass_corn_yield	USDA NASS Quick Stats	2,417	Corn grain yield (bu/acre), 102 Illinois counties, 2000–2025
02_raw_noaa_ncei_county_precipitation	NOAA NCEI	416	Monthly rainfall (inches)
03_raw_noaa_ncei_county_temperature	NOAA NCEI	416	Monthly mean temperature (°F)
04_raw_noaa_ncei_county_drought_pdsi	NOAA NCEI	416	Monthly Palmer Drought Severity Index (PDSI)

The weather files cover 16 counties (16 counties × 26 years = 416 rows): Adams, Alexander, Bond, Boone, Brown, Bureau, Calhoun, Carroll, Cass, Champaign, Christian, Clark, Clay, Clinton, Coles and Cook.

Type of dataset: Public.

Links: https://quickstats.nass.usda.gov and https://www.ncei.noaa.gov/access/monitoring/climate-at-a-glance/ (add the exact pages you downloaded from).

What PDSI means: a drought score. Negative = dry, positive = wet; −2 or lower = drought, +2 or higher = wet.

Final table: 370 records and 41 columns: county, FIPS code, year, corn yield (the target), 12 months each of rainfall, temperature and PDSI, plus 3 columns we built (growing-season temperature, rainfall and PDSI).

Honest note on size: the final table has 370 records, below the 500-record guideline, because weather was collected for only 16 of 102 counties. This size (300–400 records) matches what the approved synopsis planned. Please confirm this with your faculty mentor before ticking the "meets quality criteria" box.

3. Data Cleaning & Preprocessing

Step	What we did	Result
1. Load & inspect	Read all four CSV files	Yield: 2,417 rows × 27 columns. Each weather file: 416 rows
2. Remove useless columns	Dropped empty columns (Week Ending, Zip Code, Region, Watershed) and constant or duplicate ones (Program, Period, Geo Level, Commodity, Data Item, Domain, State, Value, noaa_element)	Kept County, fips, year, corn_yield_bu_acre, county_fips_year
3. Missing values	Only CV (%) was mostly empty (1,928 of 2,417), and we do not use it, so the column was dropped, not the rows	Weather files had no missing values
4. Duplicates	Checked the county_fips_year key in every file	No duplicates
5. Merge	Inner join of the four tables on county-year	370 matched records (46 climate county-years had no published yield)
6. Outliers	IQR rule (valid range 81.7–262.0 bu/acre) flagged 5 values: Clark 2012 (51.1), Clinton 2012 (30.2), Adams 2012 (76.0), Bond 2012 (33.3), Clay 2002	Kept, as they are real drought-year survey values

Step	What we did	Result
	(68.0)	
7. New features	gs_temp = June–Aug mean temperature; gs_precip = May–Aug total rainfall; gs_pdsi = June–Aug mean PDSI	3 new columns
8. Units & types	Rainfall in inches, temperature in °F, keys as integers	No scaling needed (we use correlations and group comparisons)

Important lesson (why we merged with an inner join and did not fill blanks with 0): filling missing values with 0 creates fake yields of 0 bu/acre and fake temperatures of 0 °F. This mistake made the early charts show a line dropping to zero and a column of points at 0 °F. The charts in this report use the corrected table.

4. Exploratory Data Analysis (EDA)

Variable (n = 370)	Mean	Median	Std dev	Min	Max
Corn yield (bu/acre)	170.4	173.6	34.0	30.2	240.8
July rainfall (in)	4.05	3.67	1.79	1.11	11.07
July mean temperature (°F)	67.5	67.65	2.63	60.8	72.7
July PDSI	0.95	1.22	2.14	−4.20	6.05

What this tells us:

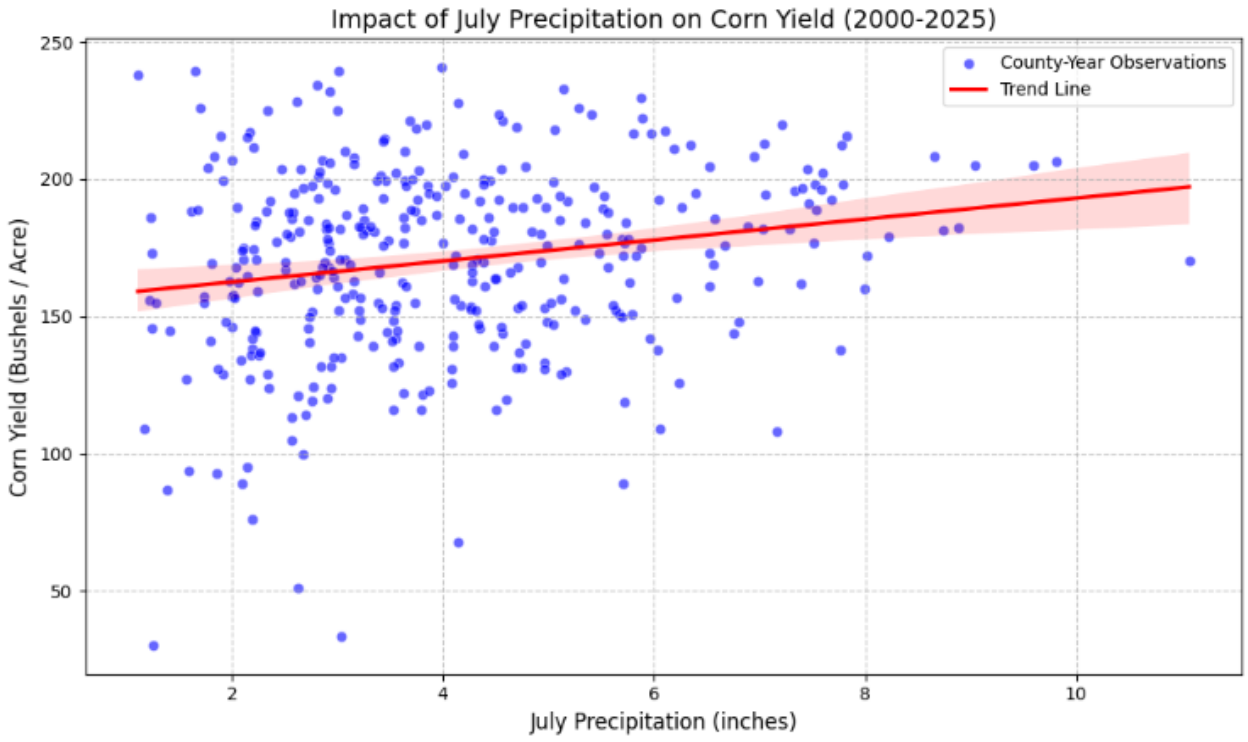
- **Yield is skewed left** (skewness −0.68): most records are high, with a few very low drought-year values pulling the average down.
- **July rainfall is skewed right**: a few very wet Julys push the mean above the median.
- **Growing season averages:** June–Aug temperature 65.1 °F (SD 1.90), May–Aug rainfall 16.2 in (SD 4.48), June–Aug PDSI 0.90 (SD 2.11).
- **Trend over time:** 152.2 bu/acre in 2000–2004 vs 205.3 in 2021–2025; linear trend about +2.7 bu/acre per year.
- **Worst and best years:** 2012 averaged 93.0 bu/acre, against 176.8 in 2013 and 209.6 in 2024.

- **Counties:** mean yield runs from 141.1 (Cook) to 197.3 (Carroll). Highest: Carroll, Bureau (189.5), Christian (189.4), Champaign (183.0), Cass (182.9), Coles (182.3). Lowest: Cook, Clay (148.6), Bond (150.7), Alexander (152.1), Clinton (153.8).
- **Between climate variables:** July rainfall and July PDSI move together (r = 0.59); July rainfall and July temperature move oppositely (r = −0.24); July temperature and PDSI are unrelated (r = 0.00).

5. Data Visualization & Observations

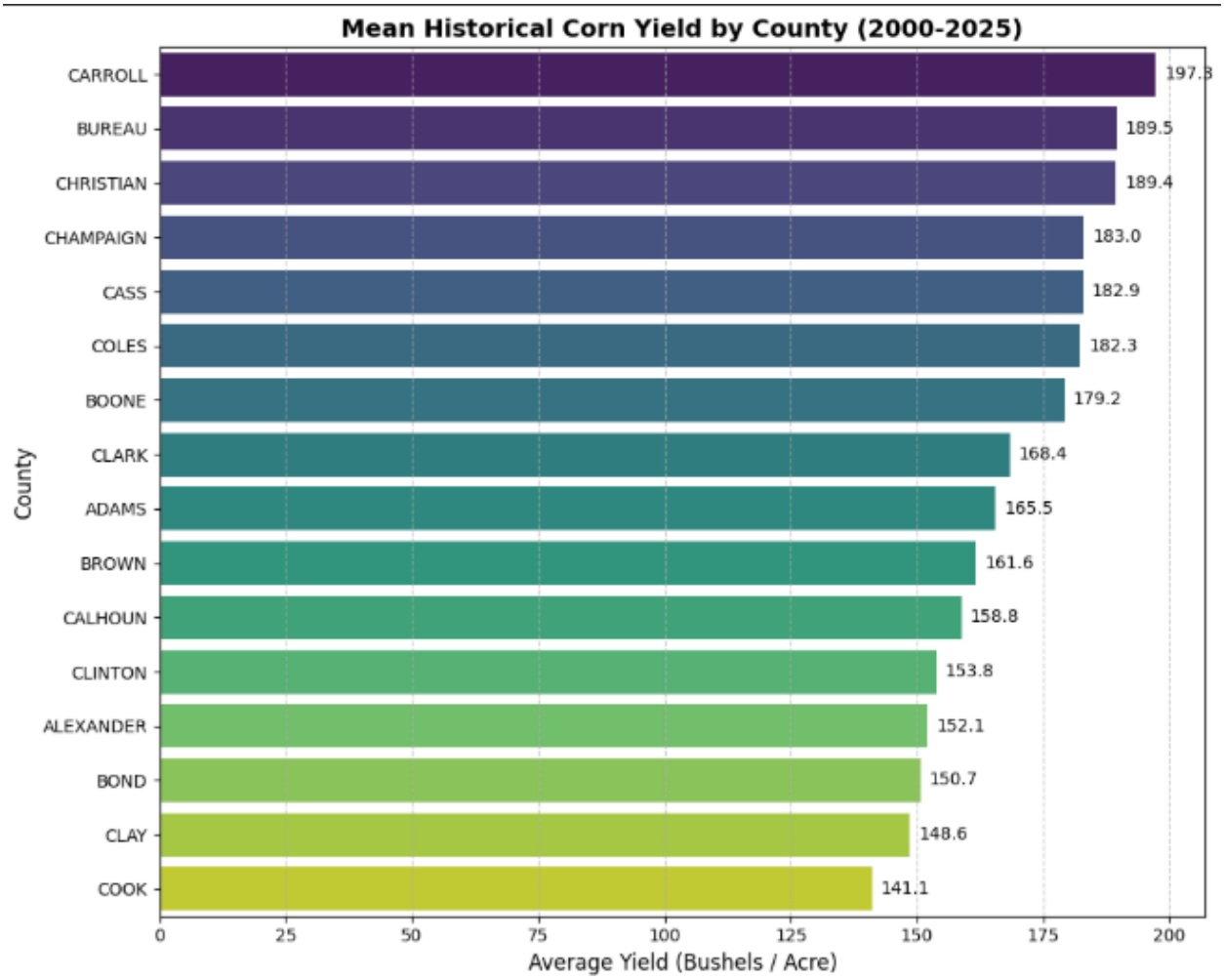
All figures use the corrected 370-row table. Each has a one-line "what it means" for non-technical readers.

Figure 1: Average corn yield by year, 2000–2025 (line chart)



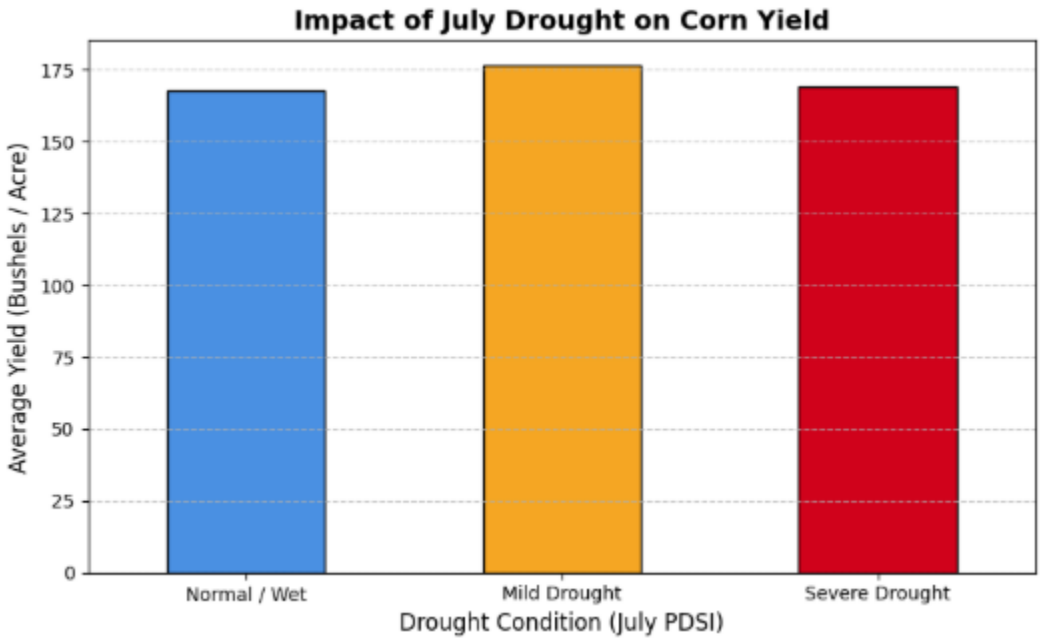
USDA NASS county corn yield, 16 Illinois counties, 2000-2025

Figure 2: Mean corn yield by county (bar chart, 16 counties)



USDA NASS county corn yield, 370 county-years, 2000-2025

Figure 3: Average yield by July drought level (bar chart)



USDA NASS yield joined with NOAA NCEI July PDSI, 370 county-years

What the figures show:

- **Figure 1:** yield climbs from about 150 bu/acre in the early 2000s to about 205 since 2021, with sharp dips in 2002 (124.1), 2005 (138.8) and above all 2012 (93.0). In 2024 and 2025 the averages rest on only 8 and 9 counties. Part of the rise is better hybrids and farm management, not climate.
- **Figure 2:** counties differ clearly, so soil, farm practice and local conditions matter alongside weather.
- **Figure 3:** the average yield for drought Julys (PDSI ≤ -2) is 168.9, mild drought 176.9 and normal/wet 167.6. There is no clear pattern here, so July PDSI alone is a weak signal (see limitations).

Table 5a: Suitability bands, yield relative to the time trend

Each yield was compared with the value expected for its year (fitted line, about 2.7 bu/acre per year). Positive = better than the trend.

Band	Records	Mean yield (bu/acre)	Yield vs. trend (bu/acre)
July temp up to 65 °F	72	167.0	+8.4
July temp 65–68 °F	130	169.1	+1.6
July temp 68–70 °F	101	169.9	−4.5
July temp above 70 °F	67	177.3	−5.4
July rainfall up to 2.5 in	70	160.4	−10.6
July rainfall 2.5–4 in	135	170.7	+0.4
July rainfall 4–6 in	113	170.7	+3.3
July rainfall above 6 in	52	182.8	+6.1

Reading the table: the raw mean yield is highest in the hottest band only because hot Julys were more common in recent, higher-yielding years. Once the trend is removed, cooler and wetter Julys perform better. The worst combination is a warm (68–70 °F) and dry (2.5 in or less) July: −20.6 bu/acre, 31 records.

6. Statistical Analysis (Basic)

Main finding: climate has a small but detectable effect on yield, and time explains more of the variation than July climate does.

Correlations with yield (n = 370). Pearson r measures a straight-line link (−1 to +1; near 0 = weak). Spearman ρ is the rank-based version. A p-value below 0.05 means the link is unlikely to be chance.

Variable	Pearson r	p-value	Spearman ρ
June rainfall	−0.249	<0.001	−0.182

Variable	Pearson r	p-value	Spearman ρ
July rainfall	0.200	0.0001	0.179
August rainfall	0.100	0.055	0.108
July mean temperature	0.093	0.073	0.130
July PDSI	−0.108	0.038	−0.123
June–Aug mean temperature	0.058	0.267	0.081
May–Aug total rainfall	−0.073	0.162	−0.057
June–Aug mean PDSI	−0.133	0.011	−0.149

Year itself correlates with yield at $r = 0.575$ ($p < 0.001$). After removing the linear time trend, the climate links become clearer: July temperature $r = -0.170$ ($p = 0.001$), July rainfall $r = +0.189$ ($p < 0.001$), June–Aug PDSI $r = -0.140$ ($p = 0.007$), June–Aug temperature $r = -0.122$ ($p = 0.019$).

Hypothesis tests (Welch t-test and ANOVA, 5% level):

Test	Group means (bu/acre)	Result
Drought July (PDSI ≤ −2, n = 29) vs other Julys (n = 341)	168.9 vs 170.5	$t = -0.27$, $p = 0.788$: no significant difference
Hotter-than-median July vs cooler July	173.7 vs 167.1	$t = 1.88$, $p = 0.061$: not significant at 5%
One-way ANOVA across 16 counties	Carroll 197.3 … Cook 141.1	$F = 7.23$, $p < 0.001$: county means differ

7. Key Insights & Interpretation

1. **Time is the biggest single factor.** Year alone explains about 33% of yield variation ($R^2 = 0.33$), about twice the 16% explained by the three July climate variables together.
2. **Extreme seasons matter more than average weather.** The 2012 county average was 93.0 bu/acre, 83.8 below 2013, and four of the five statistical outliers come from 2012.
3. **July moisture helps and July heat hurts, once the trend is removed.** Detrended yield correlates +0.19 with July rainfall and −0.17 with July temperature.

4. **A wet June goes with lower yield** ($r = -0.25$). A possible reason is waterlogged fields or delayed planting, but we did not test this.
5. **Location matters.** County means span 141.1 to 197.3 bu/acre (ANOVA $p < 0.001$).

Simple guidance: in this data the most favourable conditions are moderate July temperatures (below about 68 °F), adequate July rainfall (above about 2.5 in) and no severe summer drought. Because monthly averages explain only part of yield, use them to flag risk, not to predict exact yield.

8. Conclusion & Future Scope

In Illinois between 2000 and 2025, corn yield was driven mainly by the upward time trend (about +2.7 bu/acre per year) and by county, while monthly climate had a smaller but real effect. Cooler, wetter Julys were linked with higher yield, and warm, dry Julys with the biggest losses (about 20.6 bu/acre below trend).

Limitations

- Weather covers only 16 counties, so there are only 370 records.
- Monthly averages hide short extremes, such as heat waves during pollination, and the timing of rain.
- The PDSI column does not clearly reflect the 2012 drought: the average July PDSI for 2012 is +1.09 although yield fell to 93.0. July rainfall in 2012 was low (about 1.2–3.0 in) in most counties, which fits the drought. The PDSI extraction should be checked against the NOAA source.
- Correlation is not causation, and the yield trend also reflects better hybrids and management.
- In 2024 and 2025 only 8 and 9 counties have a published yield.

Future scope

- Collect weather for all 102 counties to pass 500 matched records and map suitability across the state.
- Add daily weather, growing degree days, soil properties and planting dates.
- Extend to other states and build a simple dashboard for county suitability.

9. References

1. USDA National Agricultural Statistics Service. Quick Stats: county corn grain yield, Illinois, 2000–2025. https://quickstats.nass.usda.gov (accessed [date]).

2. NOAA National Centers for Environmental Information. Climate at a Glance: County Time Series (precipitation, temperature, Palmer Drought Severity Index). https://www.ncei.noaa.gov/access/monitoring/climate-at-a-glance/ (accessed [date]).
3. Palmer, W. C. (1965). Meteorological Drought. Research Paper No. 45, U.S. Weather Bureau.

10. AI Tool Usage Declaration

AI tool use must be declared (PBL guidelines, Sections 4 and 15)

AI Tool	Purpose	Extent of use
Claude	Using as a reference for the understanding of the project	Used for the define the project like Format of the code and the Data Visuaization

11. Project Links & Deliverables

Item	Detail
GitHub repository	https://github.com/Mohith-ch-2008/Data-Analytics-PBL-Report
Notebook	Data_Analytics_PBL.ipynb
Report file	Corn_suitability_report.pdf

☐ GitHub repository includes code, notebook, a PBL report and README

12. Report Structure Checklist (Section 8 Compliance)

S. No.	Report Section	Included
1	Title Page	✓
2	Introduction & Problem Statement	✓
3	Dataset Description & Source	✓
4	Data Cleaning & Preprocessing	✓
5	Exploratory Data Analysis (EDA)	✓
6	Data Visualization & Observations	✓
7	Statistical Analysis (Basic)	✓
8	Key Insights & Interpretation	✓
9	Conclusion & Future Scope	✓
10	References	✓

13. Declaration by Student(s)

We hereby declare that this Mini Project Report is our own original work, prepared using publicly available datasets. All sources of data, references and AI tool assistance have been acknowledged. We understand that plagiarism in code or report will attract strict academic action, as per Section 15 of the PBL guidelines.

Team Leader Signature: _____

Team Member Signatures: _____

Date: _____

14. For Faculty Use Only – PBL Final Report Evaluation

Evaluation Component B — PBL Final Report Submission | Mid-Term (PBL Based): 10 Marks (Section 10 of PBL Guidelines)

Evaluation Criteria	Max Marks	Marks Awarded
Data Cleaning & Preprocessing	2	
Exploratory Data Analysis (EDA)	2	
Visualization & Observations	2	
Statistical Analysis	2	
Insights, Interpretation & Conclusion	2	
Total (out of 10)	**10**	

Faculty Remarks:

Faculty Mentor Signature: _____

Date: _____